

Research internship report:

Does data interpolation contradicts statistical optimality?

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1 Introduction

When designing a function estimator or statistical classifier, a trade-off is often required to achieve the lowest possible statistical risk, given a dataset $(X_i, Y_i)_{1 \leq i \leq n}$ where the X_i serve as covariates (also referred to as “features” in machine learning) and the Y_i denote the target variables (or “labels”).

Risk is the metric to be optimized and is typically given by $R(\theta, \hat{\theta}) = \mathbb{E}[(\theta - \hat{\theta})^2]$ in the case of quadratic risk. Alternative formulations of the risk may be considered (any L^p norm being suitable); however, the quadratic risk is particularly appealing because it is everywhere differentiable—facilitating gradient-based optimization algorithms—and admits a decomposition into a bias term and a variance term:

$$R(\theta, \hat{\theta}) = \mathbb{E}[(\theta - \hat{\theta})^2] = \mathbb{E}[\theta - \hat{\theta}]^2 + \text{Var}(\hat{\theta})$$

L’une des règles tacites dans l’estimation et l’apprentissage statistique est qu’il y a un compromis à réaliser entre ces deux grandeurs. En augmentant la complexité du modèle pour qu’il s’adapte parfaitement aux données qu’on lui fournit on diminue le terme de biais. Cependant sa sensibilité augmente avec sa complexité et sa variance mécaniquement aussi. Ce phénomène aussi appelé « overfitting » en apprentissage peut-être comparé à un humain qui apprendrait par coeur une chose sans essayer de reconnaître des motifs dans son apprentissage. Ce compromis qui pourrait faire penser à un principe d’incertitude de la physique quantique n’est en fait qu’une règle empirique qui peut être ignorée sous certaines hypothèses. Dans l’article de Belkin, Rakhlin et Tsybakov il est démontré que par une méthode d’interpolation spécifique, dans certains cas l’estimateur est asymptotiquement optimal au sens minimax tout en reproduisant fidèlement ses données d’entraînement.

2 The singular Nadaraya-Watson estimator

Definition 2.1 (Nadaraya-Watson estimator)

Let $K : \mathbb{R}^d \rightarrow \mathbb{R}_+$ a symmetric kernel with its support included in the unit sphere (i.e $K(x) = K(y), \forall x, y$ such that $\|x\| = \|y\|$ and $\{x \mid K(x) > 0\} \subseteq B(0, 1)$).

The Nadaraya-Watson estimator of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ associated with K is defined as such:

$$\hat{f}_n(x) = \begin{cases} Y_i & \text{if } x = X_i \text{ for some } i \in \{1, \dots, n\}, \\ 0 & \text{if } \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right) = 0, \\ \frac{\sum_{i=1}^n Y_i K\left(\frac{x - X_i}{h}\right)}{\sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)} & \text{otherwise} \end{cases}$$

This estimator can be seen as a weighted sum over all labels from the training data with the kernel images acting as the weights. The inclusion of the kernel's support serves to prevent datapoints at a distance superior to h from x (such that $\|x - X_i\| > h$) to influence the inference. h is one of the hyperparameters of this model. The **singular Nadaraya-Watson estimator** appears in the special case where K tends to infinity when approaching 0. Since

$$\lim_{x \rightarrow X_j} \frac{\sum_{i=1}^n Y_i K\left(\frac{x - X_i}{h}\right)}{\sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)} = Y_j$$

the estimator is still continuous everywhere. For the rest of this presentation and the coming simulations, the chosen singular kernel is the one used in the original article:

Definition 2.2 (The kernel K)

$$K(u) = \frac{1}{\|u\|^\alpha} \mathbf{1}_{\{\|u\| \leq 1\}}$$

With the hyperparameter α verifying $0 < \alpha < d/2$.

We denote by $\Sigma(\beta, L)$ the space of (β, L) -holder functions.

Definition 2.3 (The (β, L) -holder functions class)

A function f is said to be (β, L) -holder if:

- $\forall x, y \in \mathbb{R}^d |f(x) - f(y)| \leq L\|x - y\|^\beta$ if $\beta \in]0, 1]$
- f is continuously differentiable and $\forall x, y \in \mathbb{R}^d |f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \leq L\|x - y\|^\beta$ if $\beta \in]1, 2]$

2.1 In the case $\beta \in]0, 1]$

2.1.1 Hypothèses préliminaires

Theorem 2.1 (Asymptotic optimality in the minimax sense)

Suppose that:

1. $f(x) \in \Sigma(\beta, L_f)$ for $\beta \in]0, 1]$
2. $\forall x \in \mathbb{R}^d, E[Y \mid X = x]$ exists and $\mathbb{E}[(Y - \mathbb{E}[Y \mid X = x])^2] \leq \sigma_\xi^2 < +\infty$
3. The marginal density of X satisfies on its support $0 < p_{\min} \leq p(x) \leq p_{\max} < +\infty$

then

$$\mathcal{R}(\hat{f}_n) = \mathbb{E} \left[\left(\hat{f}_n(x) - f(x) \right)^2 \right] \leq C n^{-\frac{2\beta}{2\beta+d}}$$

with C a constant not dependent on the value of n and for $h \sim n^{-\frac{1}{2\beta+d}}$.

Condition 1 is a regularity constraint on the function. It ensures uniform continuity without requiring differentiability, thereby maintaining the utmost generality.

Condition 2 ensures that the variance of the difference between the true value of f and the error-prone measurements Y_i is finite. Consequently, the errors in Y_i remain controllable. Such measurement errors frequently occur in physics, as measuring instruments are never ideal. Therefore, it is commonly assumed that $Y_i = f(X_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{A}$ and $\mathcal{A}$ is a probability distribution with finite variance (thus satisfying Condition 2).

The final condition requires the conditionnal density to be compactly supported and to not tend to 0 at any point. This allows the support of the target function to be fully “covered” by a finite sample of data for all h when n is big enough. If the density was approaching 0 at a point x in its support the consequence would be that finding increasingly closer point to x in the dataset would become more and more difficult as the scope h of the kernel K is getting smaller.

Proof: The proof will rely on estimating $f(0)$ without loss of generality, since the problem remains invariant under a simple translation.

It will consist of upper bounding the loss by 3 terms:

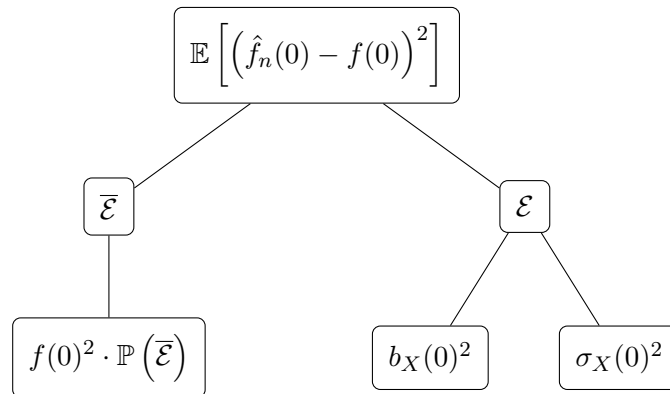


Figure 1: Les feuilles représentent les termes majorant le risque

Let's define the case

$$\mathcal{E} = \left\{ \sum_{i=1}^n K_h(X_i) \neq 0 \right\} = \{ \exists i, \|X_i\| \leq h \}$$

where one of the X_i is close enough to 0 for the estimator to compute a value.

Its complement $\bar{\mathcal{E}}$ corresponds to the scenario where the estimator fails to approximate f because the X_i are too far away. On this event, $\hat{f}_n$ vanishes, and hence the L^2 norm will be equal to $f(0)^2$. The probability of this event occurring can be bounded as follows:

$$\begin{aligned}\mathbb{P}(\bar{\mathcal{E}}) &= \mathbb{P}(\forall i \in \{1, \dots, n\}, \|X_i\| \geq h) = \mathbb{P}\left(\bigcap_{i=1}^n \{\|X_i\| \geq h\}\right) \leq \prod_{i=1}^n \mathbb{P}(\|X_i\| \geq h) \\ &= (1 - \mathbb{P}(\{\|X_1\| \leq h\}))^n\end{aligned}$$

This is where Condition 3 proves useful. It ensures that the probability of the event $\bar{\mathcal{E}}$ vanishes as the sample size increases indefinitely. This translates into the inequality $\mathbb{P}(\|X_1\| \leq h) \geq Cp_{\min}h^d$, where C is a constant independent of n reflecting the equivalence between the $\|\cdot\|_2$ and $\|\cdot\|_\infty$ norms.

Returning to $\bar{\mathcal{E}}$, this yields an exponential decay of the associated probability:

$$\mathbb{P}(\bar{\mathcal{E}}) \leq (1 - Cp_{\min}h^d)^n \leq e^{-nCp_{\min}h^d}$$

Qualitative analysis

The exponential decay can be understood from the fact that bounding the probability that $\mathcal{E}$ fails to occur as n grows yields a geometric decay in n .

The constant C reflects the equivalence of norms in finite-dimensional spaces.

The lower bound $p_{\min}$ constrains the extent of the marginal density's support: a larger value implies a more concentrated support that is covered more rapidly, thereby accelerating the convergence of the aforementioned bound to 0.

The final term h^d in this exponential reflects the necessity of scaling the radius h of the balls with respect to the dimension d of the features space. Since h must vanish asymptotically, it will eventually satisfy $h < 1$. Under this condition, increasing the dimension d decreases h^d , thus requiring a larger bandwidth h in higher dimensions. Nevertheless, for any fixed dimension d , this exponential bound remains asymptotically negligible.

Now that the loss has been bounded on the event $\bar{\mathcal{E}}$ let's focus on the opposite case. Conditionally on $\mathcal{E}$,

$$\hat{f}_n(0) = \frac{\sum_{i=1}^n Y_i K_h(X_i)}{\sum_{i=1}^n K_h(X_i)}.$$

Proposition 2.1 (Bias-variance decomposition)

$$\begin{aligned}\mathbb{E}\left[\left(\hat{f}_n(0) - f(0)\right)^2\right] &\leq \mathbb{E}\left[\left(\hat{f}_n(0) - f(0)\right)^2 \mathbf{1}(\mathcal{E})\right] + \mathbb{P}(\bar{\mathcal{E}}) \\ &= \mathbb{E}\left[\left(\hat{f}_n(0) + \mathbb{E}\left[\hat{f}_n(0) \mid X\right] - \mathbb{E}\left[\hat{f}_n(0) \mid X\right] - f(0)\right)^2 \mathbf{1}(\mathcal{E})\right] + \mathbb{P}(\bar{\mathcal{E}}) \\ &= \underbrace{\mathbb{E}\left[\left(\hat{f}_n(0) - \mathbb{E}\left[\hat{f}_n(0) \mid X\right]\right)^2 \mathbf{1}(\mathcal{E})\right]}_{\text{Variance term } \sigma_X^2(0)} + \underbrace{\mathbb{E}\left[\left(\mathbb{E}\left[\hat{f}_n(0) \mid X\right] - f(0)\right)^2 \mathbf{1}(\mathcal{E})\right]}_{\text{Bias term } b_X^2(0)} + \mathbb{P}(\bar{\mathcal{E}})\end{aligned}$$

Where we used the previous result to obtain the first inequality. To get the last equality, implicitly:

$$\begin{aligned} & \mathbb{E} \left[\left(\hat{f}_n(0) + \mathbb{E} [\hat{f}_n(0) | X] - \mathbb{E} [\hat{f}_n(0) | X] - f(0) \right)^2 \mathbf{1}(\mathcal{E}) \right] \\ &= \mathbb{E} [(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) | X])^2 \mathbf{1}(\mathcal{E})] + \mathbb{E}[(\mathbb{E}[\hat{f}_n(0) | X] - f(0))^2 \mathbf{1}(\mathcal{E})] \\ & \quad + 2\mathbb{E}[(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) | X])(\mathbb{E}[\hat{f}_n(0) | X] - f(0))] \end{aligned}$$

and

$$\begin{aligned} & \mathbb{E} [(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) | X])(\mathbb{E}[\hat{f}_n(0) | X] - f(0))] \\ & \leq \underbrace{\mathbb{E} [(\hat{f}_n(0) - \mathbb{E}[\hat{f}_n(0) | X])]}_{=0} \mathbb{E} [\mathbb{E}[\hat{f}_n(0) | X] - f(0)] \end{aligned}$$

The rest of the demonstration will consist in bounding these two terms by using the hypotheses.

2.1.2 Bounding of the variance term

$$\begin{aligned} \sigma(0)^2 &= \mathbb{E} \left[\left(\frac{\sum_{i=1}^n \xi_i K_h(X_i)}{\sum_{i=1}^n K_h(X_i)} \right)^2 \mathbf{1}(\mathcal{E}) \right] \leq \mathbb{E} \left[\left(\max_{1 \leq i \leq j} \xi_i \right)^2 \right] \mathbb{E} \left[\frac{\sum_{i=1}^n K_h(X_i)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \right] \\ &\leq \sigma_\xi^2 \cdot n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \right] \end{aligned}$$

By denoting σ_X the terme

$$n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\mathcal{E}) \right]$$

we aim to show that its convergence is polynomial.

Let's denote

$$\mathcal{A} = \left\{ \sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right\}.$$

The bound $\frac{n\bar{p}}{2}$ hasn't been chosen randomly, it will allow us to use Bernstein ineaquility later.

By partitioning the space with $\mathcal{A}$:

$$\sigma_X^2 \leq n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\mathcal{A} \cap \mathcal{E}) \right] + n \mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i) \right)^2} \mathbf{1}(\bar{\mathcal{A}} \cap \mathcal{E}) \right] \quad (1)$$

If $\|X_i\| \leq h$, then $K_h(X_i) \geq 1 = \eta_i$; otherwise, $0 = \eta_i \leq K_h(X_i)$. Hence,

$$\sum_{i=1}^n K_h(X_i) \geq \sum_{i=1}^n \eta_i.$$

The second term can be upper bounded by:

$$n\mathbb{E} \left[\frac{4}{(n\bar{p})^2} K_h(X_1)^2 \right]$$

The kernel K belongs to L^2 . By radial symmetry and a spherical change of variables in $\mathbb{R}^d$,

$$\mathbb{E} [K_h(X_1)^2] \leq h^d p_{\max} \int_{\mathbb{R}^d} K^2(u) du \leq h^d p_{\min} C_K \int_0^1 r^{-2a} r^{d-1} dr.$$

Choosing $a < \frac{d}{2}$ ensures that $d - 1 - 2a < 1$, thereby establishing the finiteness of $\mathbb{E} [K_h(X_1)^2]$. Consequently, the second term converges at a rate of $O\left(\frac{1}{n}\right)$.

Returning to the term

$$n\mathbb{E} \left[\frac{K_h(X_1)^2}{\left(\sum_{i=1}^n K_h(X_i)\right)^2} \mathbf{1}(\mathcal{A} \cap \mathcal{E}) \right],$$

it is upper bounded by

$$n\mathbb{P} \left(\sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right) \quad \text{since} \quad K_h(X_1)^2 \leq \left(\sum_{i=1}^n K_h(X_i) \right)^2.$$

The events

$$\left\{ \sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right\} \quad \text{and} \quad \left\{ \sum_{i=1}^n \eta_i - n\bar{p} \leq \frac{n\bar{p}}{2} \right\}$$

are not identical; however, their probabilities coincide because the variables η_i are independent Bernoulli variables with parameter $\bar{p}$. The probability of this latter event can be bounded using Bernstein's inequality, as the variables $|\eta_i - \bar{p}|$ are zero-mean and bounded almost surely by 1.

$$\begin{aligned} n\mathbb{P} \left(\sum_{i=1}^n \eta_i \leq \frac{n\bar{p}}{2} \right) &\leq 2n \exp \left(\frac{-\left(\frac{n\bar{p}}{2}\right)^2}{2 \sum_{i=1}^n \mathbb{E}[(\eta_i - \bar{p})^2] + \frac{2}{3}n\bar{p}} \right) = 2n \exp \left(-\frac{n\bar{p}}{8((1 - \bar{p}) + \frac{1}{3})} \right) \\ &\leq 2n \exp \left(\frac{n\bar{p}}{8 \cdot \frac{4}{3}} \right) = 2n \exp \left(\frac{3n\bar{p}}{32} \right) \end{aligned}$$

This concludes the discussion on the convergence of the variance term.

2.1.3 Bounding the Bias Term

Exploiting the property that $f_X \in \Sigma(\beta, L_f)$,

$$\left| \frac{\sum_{k=1}^n (f(X_k) - f(0))^2 K_h(X_k)}{\sum_{k=1}^n K_h(X_k)} \right| \leq \left| \frac{\sum_{k=1}^n L_f \|X_k\|^\beta K_h(X_k)}{\sum_{k=1}^n K_h(X_k)} \right| \leq L_f h^\beta$$

Thus, $b^2(0) \leq L_f^2 h^{2\beta}$.

2.2 Conclusion

Combining all previous upper bounds yields:

$$\mathbb{E} \left[\left(\hat{f}_n(X) - f(0) \right)^2 \right] \leq n \exp(-Cnh^d) + \frac{C}{nh^d} + Ch^{2\beta}$$

The first term can be neglected as it is asymptotically dominated by any polynomial decay. Balancing the last two terms yields the optimal bandwidth and the associated risk:

$$h_n \sim n^{-\frac{1}{2\beta+d}}$$

$$\mathbb{E} \left[\left(\hat{f}_n(0) - f(0) \right)^2 \right] \leq Cn^{-\frac{2\beta}{2\beta+d}}$$

3 Extension of the Result to $\beta \in]1, 2]$

$f \in \Sigma(\beta, L)$ with $\beta \in]1, 2]$ means that $\forall x, y \in \mathbb{R}^d, |f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \leq L\|x - y\|^\beta$. In this setting, f must be continuously differentiable. The associated smoothness constraint is rather stringent, as evidenced by these two figures:

Assuming in addition that $p_X \in \Sigma(\beta - 1, L_p)$, we obtain

$$b^2(0) \leq (L_f + \|\nabla f(0)\| L_p p_{\min}^{-1}) h^{2\beta} + \sigma_X^2$$

Setting

$$G_i = \frac{(f(X_i) - f(0))K_h(X_i)}{\sum_{j=1}^n K_h(X_j)},$$

we obtain the following formulation of $b^2(0)$:

$$b^2(0) = \mathbb{E} \left[\sum_{i,j}^n G_i G_j \mathbf{1}(\mathcal{E}) \right].$$

For $i \neq j$, the expectation of the product can be written as:

$$\mathbb{E}[G_i G_j I\{\mathcal{E}\}] = \mathbb{E}[(f(X_i) - f(0))(f(X_j) - f(0))A(X_i, X_j)] \quad (2)$$

with:

$$A(X_i, X_j) = \frac{K_h(X_i)K_h(X_j)}{(\sum_{i=1}^n K_h(X_i))^2} I\{\mathcal{E}\} \geq 0.$$

For conciseness, we omit the explicit dependence of $A(X_i, X_j)$ on the variables $(X_k, k \neq i, k \neq j)$. This leads to:

$$\mathbb{E}'[G_i G_j I\{\mathcal{E}\}] = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (f(x_i) - f(0))(f(x_j) - f(0))A(x_i, x_j)p(x_i)p(x_j)dx_i dx_j \quad (3)$$

where $\mathbb{E}'$ denotes the conditional expectation given (X_i, X_j) with $(X_k, k \neq i, k \neq j)$ held fixed.

Let us define the first-order Taylor remainder:

$$R(x_i) = f(x_i) - f(0) - \langle \nabla f(0), x_i \rangle$$

$$R(x_j) = f(x_j) - f(0) - \langle \nabla f(0), x_j \rangle$$

We have

$$\begin{aligned} (f(x_i) - f(0))(f(x_j) - f(0)) &= (R(x_i) + \langle \nabla f(0), x_i \rangle)(R(x_j) + \langle \nabla f(0), x_j \rangle) \\ &= R(x_i)R(x_j) + R(x_i)\langle \nabla f(0), x_j \rangle + R(x_j)\langle \nabla f(0), x_i \rangle + \langle \nabla f(0), x_i \rangle \langle \nabla f(0), x_j \rangle \end{aligned}$$

Substituting this expansion into the integral yields three terms, since $R(x_i)\langle \nabla f(0), x_j \rangle$ is symmetric under permutation of x_i and x_j , leaving the integral unchanged:

$$\begin{aligned} \mathbb{E}'[G_i G_j I\{\mathcal{E}\}] &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle \langle \nabla f(0), x_j \rangle A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j \\ &\quad + 2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle R(x_j) A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j \\ &\quad + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} R(x_i) R(x_j) A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j \end{aligned}$$

Since the inner product is odd and $A(x_i, x_j)$ is even, it follows that:

$$\int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle A(x_i, x_j) p(0) dx_i = 0$$

for all x_j , because the integrand is odd with respect to x_i for any fixed $(X_k, k \neq i, k \neq j)$. Applying this observation to x_i and x_j in the first term, as well as in the second term, the decomposition becomes:

$$\begin{aligned} \mathbb{E}'[G_i G_j I\{\mathcal{E}\}] &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle \langle \nabla f(0), x_j \rangle A(x_i, x_j) (p(x_i) - p(0))(p(x_j) - p(0)) dx_i dx_j \\ &\quad + 2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \langle \nabla f(0), x_i \rangle R(x_j) A(x_i, x_j) (p(x_i) - p(0)) p(x_j) dx_i dx_j \\ &\quad + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} R(x_i) R(x_j) A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j \end{aligned} \tag{4}$$

The Hölder class condition implies that $|R(x_i)| \leq L_f \|x_i\|^\beta$. Moreover, $A(x_i, x_j)$ vanishes whenever $\|x_i\| > h$ or $\|x_j\| > h$. Applying the Cauchy-Schwarz inequality to the inner products along with the Hölder assumption on p , we obtain the following bound:

$$\begin{aligned} \mathbb{E}'[G_i G_j I\{\mathcal{E}\}] &\leq B^2 L_p^2 h^{2\beta} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} A(x_i, x_j) dx_i dx_j \\ &\quad + 2 B L_f L_p h^{2\beta} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} A(x_i, x_j) p(x_j) dx_i dx_j \\ &\quad + L_f^2 h^{2\beta} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} A(x_i, x_j) p(x_i) p(x_j) dx_i dx_j \end{aligned}$$

where $B = \|\nabla f(0)\|^2$.

Factoring out common terms, exploiting the lower bound $p_{\min}$ of the density, and taking the expectation over the remaining variables $(X_k, k \neq i, k \neq j)$, we establish:

$$\mathbb{E}[G_i G_j I\{\mathcal{E}\}] \leq h^{2\beta} \left(B L_p p_{\min}^{-1} + L_f \right)^2 \mathbb{E} \left[\frac{K_h(X_i) K_h(X_j)}{(\sum_{i=1}^n K_h(X_i))^2} I\{\mathcal{E}\} \right] \tag{5}$$

When $j = i$, we obtain the variance term:

$$\sum_{i=1}^n \mathbb{E}[G_i^2 I\{\mathcal{E}\}] = n \mathbb{E} \left[\frac{K^2(X_1)}{(\sum_{i=1}^n K_h(X_i))^2} I\{\mathcal{E}\} \right] = \sigma_X^2 \tag{6}$$

Finally, for the cross-product terms:

$$\begin{aligned}
\sum_{i \neq j} \mathbb{E}[G_i G_j I\{\mathcal{E}\}] &= h^{2\beta} \left(BL_p p_{\min}^{-1} + L_f \right)^2 \mathbb{E} \left[\frac{\sum_{i \neq j} K_h(X_i) K_h(X_j)}{(\sum_{i=1}^n K_h(X_i))^2} I\{\mathcal{E}\} \right] \\
&\leq h^{2\beta} \left(BL_p p_{\min}^{-1} + L_f \right)^2 \mathbb{E} \left[\frac{\sum_{i,j=1}^n K_h(X_i) K_h(X_j)}{(\sum_{i=1}^n K_h(X_i))^2} I\{\mathcal{E}\} \right] \\
&\leq h^{2\beta} \left(BL_p p_{\min}^{-1} + L_f \right)^2
\end{aligned}$$

4 Explanation of the Breakthrough

This result, counter-intuitive in light of classical intuition, can be explained by two main arguments.

Interpolation has historically raised concerns regarding the inflation of the variance term. Returning to inequality (1), the two arguments proceed as follows.

The first argument applies to the second term. Assuming $a/2 \leq d$, the kernel is square-integrable (L^2). In terms of this norm, the kernel values around the singularity are sufficiently small so that, asymptotically, the effect of interpolation around this point becomes increasingly localized. As more kernels are aggregated in this model, their individual influence diminishes.

The second argument consists in upper bounding the probability that the origin 0 lies abnormally far from too many sample points $(X_i)_{i=1}^n$. Since this event is exponentially unlikely (by Bernstein's inequality), one can deduce that only the behavior of the estimator when 0 is sufficiently close to the sample points X_i governs the quality of the estimation (the phrase *sufficiently close* meaning that 0 is within distance less than 1 of at least $n\bar{p}/2$ distinct points X_i).

5 Simulations

In this section, we illustrate the empirical performance of the singular Nadaraya-Watson estimator via numerical simulations on a binary classification task.

5.1 Experimental Setting

We consider a binary classification setting where the target variable Y takes values in $\{-1, 1\}$. We investigate the behavior of the estimator under two distinct configurations for the distribution of the covariates X :

1. **Gaussian Model:** The class-conditional densities, denoted by $f_{Y=-1}(X)$ and $f_{Y=1}(X)$, are Gaussian densities with respective means $\mu_{-1} = -(0.5, 0.5)$ and $\mu_1 = (0.5, 0.5)$. They share a common covariance matrix Σ . To evaluate the robustness of the model to label noise and class overlap, three covariance regimes are evaluated:
 - $\Sigma = \text{Id}$ (baseline model),
 - $\Sigma = 1.2 \cdot \text{Id}$ (intermediate variance),
 - $\Sigma = 2 \cdot \text{Id}$ (high variance).
2. **Nonlinear Model:** To probe the adaptability of the method when faced with geometrically more complex decision boundaries, a nonlinear setting is considered. The prior conditional probability is defined by:

$$\mathbb{P}(Y = -1 \mid X) = f(X) \quad \text{with} \quad f(X) = \frac{-\frac{X_1^2}{16} + 0.5}{0.833}$$

5.2 Classification Method

To separate the two classes, the Bayes decision rule consists in evaluating the sign of the regression function $\mathbb{E}[Y | X]$. In practice, we evaluate the singular Nadaraya-Watson estimator constructed using a kernel that exhibits a singularity at the origin. The predicted label $\hat{Y}$ at a query point x is directly obtained by thresholding the estimator:

$$\hat{Y} = \text{sign}(\hat{f}_n(x))$$

Although the estimator strictly interpolates the training sample, tuning the bandwidth alongside a proper singularity exponent $\alpha < d/2$ ensures that the variance remains asymptotically controlled, thereby yielding a stable decision boundary at the macro scale.

5.3 Results and Analysis

Figures 2, 3, and 4 depict the decision boundaries obtained for the Gaussian model across the three variance settings under study.

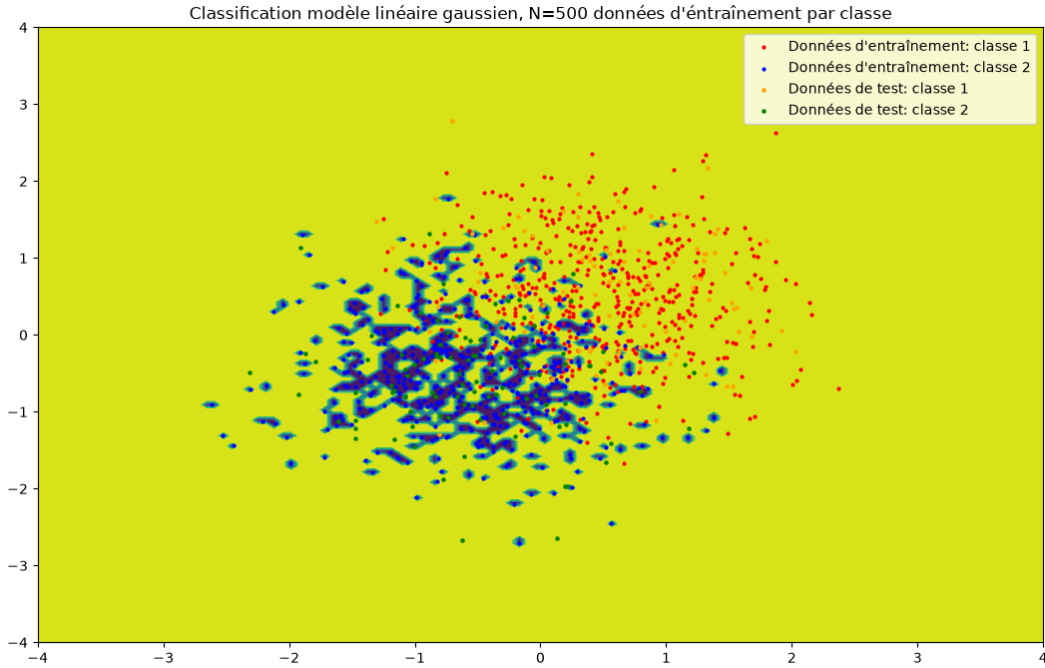


Figure 2: Classification under the Gaussian model with baseline variance ($\mu_1 = (0.5, 0.5), \mu_{-1} = -(0.5, 0.5), \Sigma = \text{Id}$).

Inspecting these initial figures indicates that despite increasing class overlap as the variance grows, the estimator preserves a globally consistent separation. The interpolation induced by the kernel singularity acts locally around the training samples without triggering macroscopic oscillations or catastrophic overfitting, visually validating the theoretical risk bounds.

Lastly, Figure 5 displays the classification boundaries obtained under the nonlinear scenario.

In this setting, the resulting decision boundary deviates locally from the theoretical parabolic expectation. This highlights the capacity of the singular Nadaraya-Watson estimator to adapt to complex geometric structures through local flexibility while preventing global overfitting.

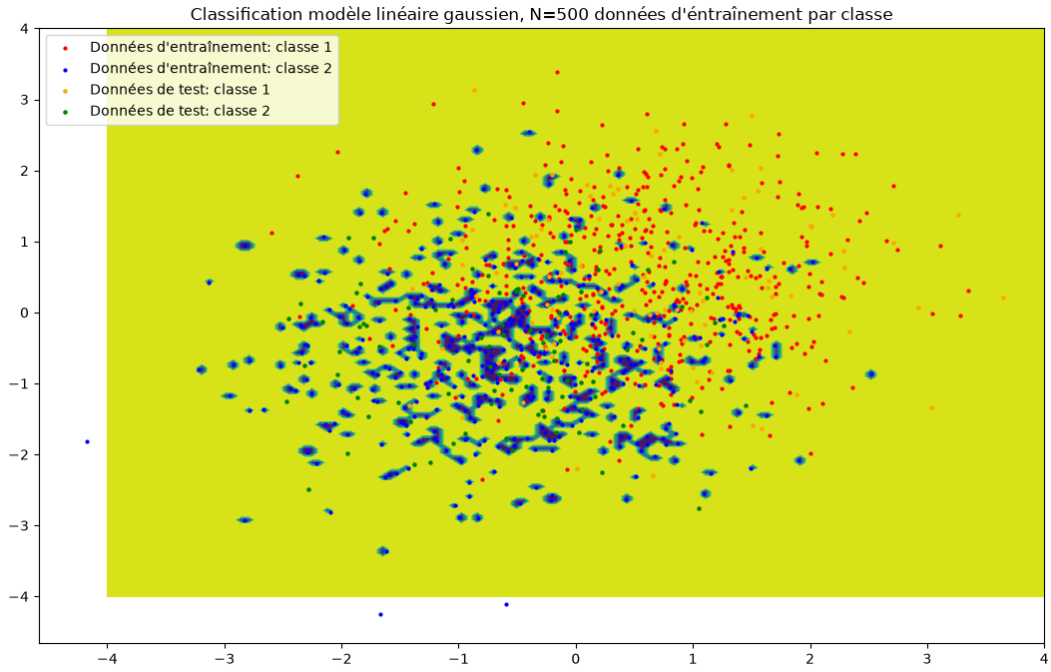


Figure 3: Classification under the Gaussian model with intermediate variance ($\mu_1 = (0.5, 0.5), \mu_{-1} = -(0.5, 0.5), \Sigma = 1.2 \cdot \text{Id}$).

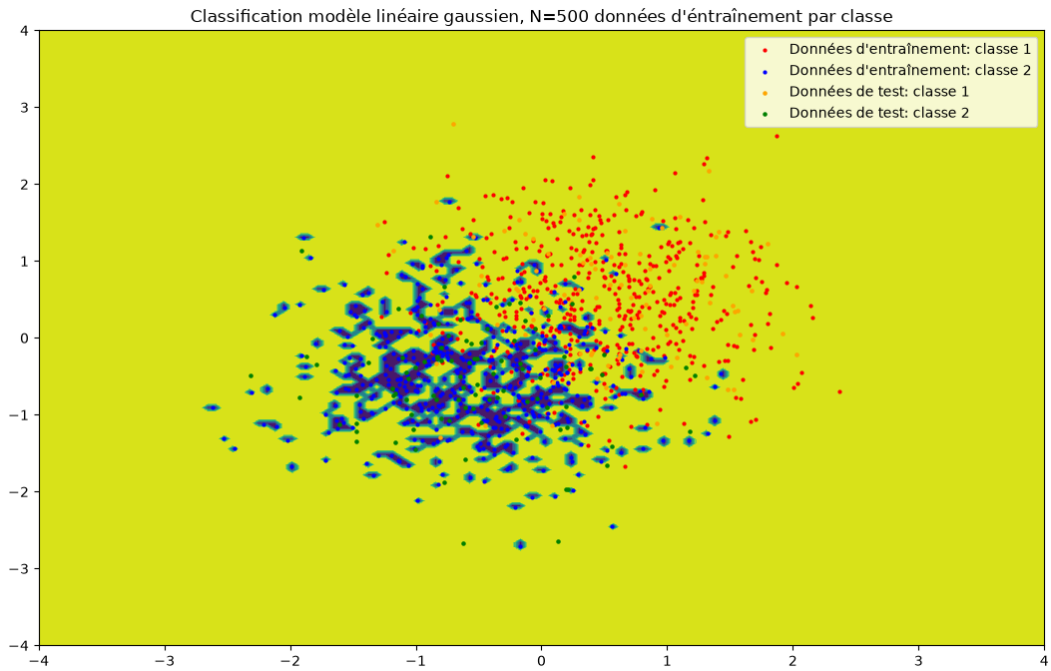


Figure 4: Classification under the Gaussian model with high variance ($\mu_1 = (0.5, 0.5), \mu_{-1} = -(0.5, 0.5), \Sigma = 2 \cdot \text{Id}$).

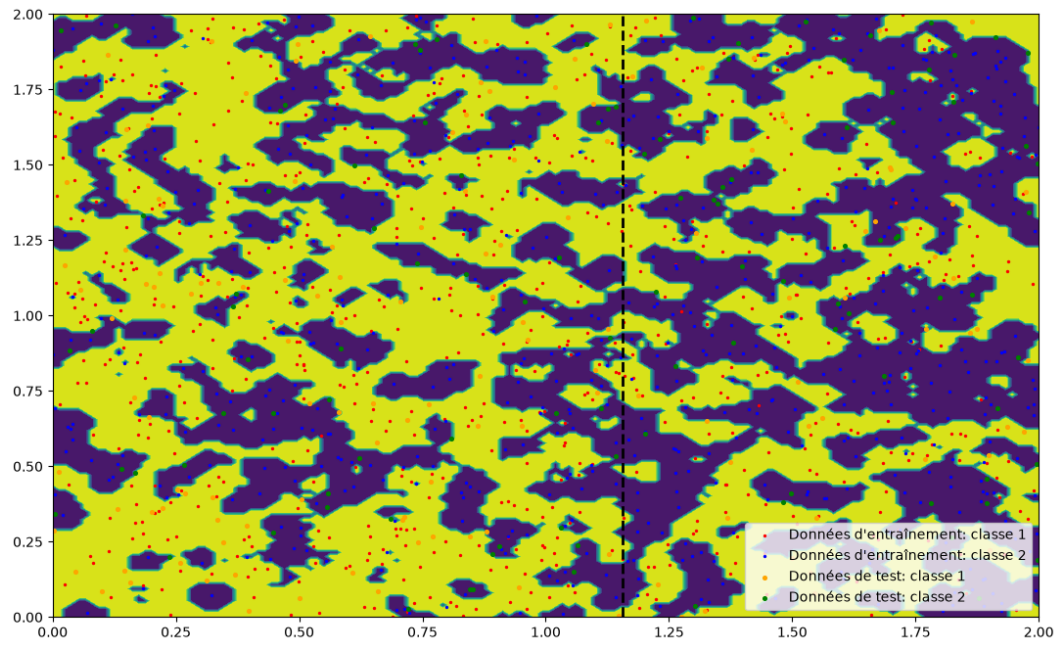


Figure 5: Classification under the nonlinear model with conditional distribution $\mathbb{P}(Y = -1 \mid X) = \frac{-X_1^2/16+0.5}{0.833}$.